

PRD: Kos Management SaaS

Working name: KosKu (placeholder) · **Version:** 0.3 draft · **Platform:** Android first, iOS later · **Market:** Indonesia, launching in Jabodetabek · **Team:** solo founder · **Model:** SaaS (owner subscription + tenant transaction fee)

1. Overview

Problem. Kos owners run their business with notebooks, WhatsApp, and memory. They lose track of who has paid, chase late tenants manually, check bank transfers one by one, and have no clear view of income or vacancies.

Solution. A simple smartphone app where the owner manages rooms, tenants, billing, and payments. Tenants need no app: they receive invoices, reminders, and receipts via WhatsApp and pay through a link. It is delivered as a multi-tenant SaaS: each owner subscribes to a plan and all data lives in the cloud under their own account.

Vision. The default operating tool for small and medium kos owners in Indonesia, with payment collection as the core value.

Goals

- Owner can set up a property and send the first invoice in under 10 minutes.
- Cut owner time spent on rent collection by at least 70%.
- Raise on-time payment rate through automated reminders and easy online payment.
- Build recurring revenue: owners subscribe monthly or annually, plus a convenience fee paid by tenants who pay online.

Non-goals (v1)

- Tenant marketplace or room listing (Mamikos territory).
 - Full accounting, payroll, or tax features.
 - Smart-lock or IoT integration.
 - A dedicated tenant app.
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2. Users

Persona	Description	Needs
Owner (primary)	5-50 rooms, age 35-65, moderate smartphone skill, WhatsApp-heavy	Know who paid, get paid on time, see income, minimal effort
Manager / staff	Employee or family member who handles daily operations	Record payments, check in tenants, limited access
Tenant	Student or worker, 18-35	Clear invoice, easy payment, proof of

3. Business Flow

3.1 Owner lifecycle (business side)

1. **Acquire:** owner communities, Facebook groups, campus/industrial-area canvassing, owner referrals.
2. **Onboard:** sign up with phone number + OTP, add property, add rooms, import tenants.
3. **Activate:** first invoice sent within 7 days.
4. **Monetize:** free trial, then a paid subscription by room count, plus a convenience fee paid by tenants who pay online.
5. **Retain:** monthly income summary, referral reward, owner support via WhatsApp.
6. **Renew / expand:** monthly or annual renewal, upgrade when room count grows, downgrade or cancel in-app.

3.2 Core monthly loop

1. System generates invoice per tenant (rent + extras) before due date.
2. Invoice sent to tenant via WhatsApp with payment link.
3. Reminders: H-3, due day, H+1, H+3 (configurable).
4. Tenant pays via VA / QRIS / e-wallet, or pays cash/transfer to owner.
5. Online payments auto-reconcile; manual payments are marked paid by owner (with optional proof photo).
6. Receipt sent to tenant automatically; owner's dashboard updates.
7. Funds settle to owner's bank account. The convenience fee is paid by the tenant at checkout, not deducted from the owner.
8. Monthly report: income, outstanding, occupancy.

3.3 Tenant lifecycle

- **Check-in:** add tenant, assign room, set rent, billing cycle, deposit, start date → first invoice (prorated if needed).
- **During stay:** monthly billing, extra charges (electricity, water, laundry, parking), complaints.
- **Contract expiry:** alert owner 30 days before; renew or end.
- **Check-out:** final invoice, deposit refund calculation, room set to vacant.

4. Scope

Phase	Contents
MVP (Phase 1)	Auth, property/room/tenant management, invoicing, manual payment recording, WhatsApp invoice + reminders + receipts, dashboard, basic report, free trial with plan limits (subscription activated manually for pilot owners), basic admin console

- Phase 2** Payment gateway (VA, QRIS, e-wallet), auto-reconciliation, settlement, late fees, subscription billing and renewals, tenant convenience fee
- Phase 3** Multi-property, staff roles, complaints/maintenance, contract renewal, export PDF/Excel, self-serve plan changes, referral program, tenant app (optional)
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5. Functional Requirements

Priority: **P0** = must have for that phase, **P1** = should have, **P2** = nice to have.

5.1 Account & access

ID	Requirement	Priority
F-01	Sign up/login with phone number + OTP (no password)	P0
F-02	Owner profile and bank account for settlement	P0 (Phase 2)
F-03	Staff accounts with limited roles (record payment, view only)	P1

5.2 Property, rooms, tenants

ID	Requirement	Priority
F-10	Create property with name and address	P0
F-11	Add rooms (name/number, type, monthly price, facilities, status: vacant / occupied)	P0
F-12	Bulk-add rooms (e.g. "create rooms 1-20 at Rp X")	P0
F-13	Add tenant: name, WhatsApp number, room, move-in date, rent, billing cycle, deposit	P0
F-14	Tenant ID photo (KTP) upload is optional and stored encrypted	P1
F-15	Check-out flow with deposit calculation	P1
F-16	Multiple properties under one account	P1

5.3 Billing

ID	Requirement	Priority
F-20	Auto-generate recurring invoices (monthly; support weekly, quarterly, yearly)	P0
F-21	Due date either fixed (e.g. every 1st) or per tenant move-in date	P0
F-22	Extra charges per invoice (electricity, water, laundry, parking, custom)	P0

F-23	Prorated first month	P1
F-24	Late fee rules (fixed or percentage, after grace period)	P1
F-25	Deposit tracking (received, deducted, refunded)	P1
F-26	Owner-selectable deposit policy per property: (a) record only, owner holds the deposit offline; (b) tenant pays deposit via the app and it settles to the owner; (c) deduct from the last month's rent; (d) deduct damages/fees at check-out and record the refund. The platform never holds deposit funds	P1

5.4 Payments

ID	Requirement	Priority
F-30	Owner marks invoice paid (cash/transfer), with date, method, optional proof photo	P0
F-31	Partial payments	P1
F-32	Online payment via payment link: virtual account, QRIS, e-wallet through a licensed gateway	P0 (Phase 2)
F-33	Auto-reconciliation of online payments via gateway webhook	P0 (Phase 2)
F-34	Settlement to owner's bank account	P0 (Phase 2)
F-36	Tenant sees an itemized convenience fee before paying online; paying the owner directly stays free; owner can choose whether to offer online payment per property	P0 (Phase 2)
F-35	Receipt generated automatically (shareable image/PDF)	P0

5.5 Notifications (WhatsApp)

ID	Requirement	Priority
F-40	Send invoice, reminder, and receipt to tenant via WhatsApp	P0
F-41	Configurable reminder schedule per property	P1
F-42	Owner push notification when tenant pays or becomes overdue	P0
F-43	Fallback: one-tap "share to WhatsApp" if API send fails	P0

5.6 Dashboard & reports

ID	Requirement	Priority
F-50	Home: paid / unpaid / overdue counts and amounts for the current month, vacant rooms	P0
F-51	Tenant list with status filters	P0
F-52	Monthly income report (collected vs outstanding), occupancy rate	P1
F-53	Export report to PDF/Excel	P2

5.7 Operations

ID	Requirement	Priority
F-60	Complaint/maintenance tickets (tenant reports via WhatsApp link, owner tracks status)	P1
F-61	Contract expiry alerts and renewal	P1
F-62	Announcements to all tenants	P2

5.8 SaaS & subscription

ID	Requirement	Priority
F-70	Multi-tenant accounts: each owner is an isolated account; all data scoped to it	P0
F-71	Free trial (proposed 30 days) from sign-up, no payment details needed	P0
F-72	Plans with entitlements: max rooms, WhatsApp message quota, features (multi-property, staff, reports)	P0
F-73	Plan limit enforcement with friendly upgrade prompts; never delete or lock existing data	P0
F-74	Subscription billing, monthly and annual, paid via VA / QRIS / e-wallet payment link, with invoice and receipt to the owner	P0 (Phase 2)
F-75	Renewal reminders to the owner via WhatsApp; grace period, then read-only mode if unpaid (data kept)	P0 (Phase 2)

F- 76	Self-serve upgrade, downgrade, cancel; owner can export own data on cancel	P1
F- 77	Internal admin console: account list, plan and trial management, usage, support lookup, audit log	P0 (basic in Phase 1)
F- 78	Referral codes with reward (e.g. one free month)	P2
F- 79	In-app support entry (WhatsApp chat to support) and short help content	P1
F- 80	SaaS metrics dashboard: MRR, churn, trial conversion, GMV, cost per account	P1

6. Non-Functional Requirements

- **Usability:** simple Bahasa Indonesia, large tap targets, max 3 taps for core actions (mark paid, send reminder). Tested with real owners aged 40+.
 - **Performance:** runs smoothly on low-end Android (2-3 GB RAM), usable on weak 3G/4G.
 - **Offline:** recording payments and viewing tenant list works offline; syncs when online.
 - **Reliability:** invoice generation and reminders run on schedule with retry; payment webhooks idempotent.
 - **Security:** encrypted data in transit and at rest; role-based access; audit log of payment changes.
 - **Privacy / compliance:** comply with Indonesia's Personal Data Protection Law (UU PDP): clear consent, tenant opt-in for WhatsApp messages, data deletion on request, minimal collection of ID data. Use only licensed payment providers; do not hold funds directly.
 - **Availability target:** 99.5% for core services.
 - **Multi-tenancy:** every record scoped to an owner account; no cross-account data access; tested isolation; per-account backups and export.
 - **Unit economics:** track cost per active owner (hosting + WhatsApp messages) against subscription revenue; enforce per-plan message quotas.
 - **Scalability:** design for thousands of accounts on one deployment; scheduled jobs (invoices, reminders, renewals) must scale without per-owner manual work.
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7. Key Data Entities

Account (owner) → Property → Room → Tenancy (tenant, room, rent, cycle, deposit, dates) → Invoice (items, due date, status) → Payment (method, amount, gateway reference, proof) · plus Tenant, Staff, Reminder/MessageLog, Complaint, Settlement, and for SaaS: Plan, Subscription, SubscriptionInvoice, UsageCounter.

8. Integrations

Need	Options
Payments	Midtrans, Xendit, Duitku (VA, QRIS, e-wallet)
Subscription billing	Gateway payment links plus your own subscription logic (simplest for solo); recurring auto-debit later if supported
WhatsApp	Official WhatsApp Business API via an authorized provider. Avoid unofficial gateways: they risk number bans and unreliable delivery
OTP	WhatsApp or SMS provider
Push	Firebase Cloud Messaging
Crash/analytics	Firebase Crashlytics, Mixpanel or PostHog

Suggested stack: Flutter or React Native (app), Node.js/Go/Laravel backend, PostgreSQL, a job queue for scheduled invoices and reminders, cloud hosting in Indonesia or Singapore.

9. Monetization (SaaS)

Two revenue streams:

1. **Owner subscription** for core features, priced by number of rooms.
2. **Tenant convenience fee**, only when the tenant pays through the app (pay-if-you-use). Paying the owner directly stays free.

Plan hypotheses (to test in the pilot, not final)

Plan	Rooms	Price / month (idea)	Includes
Trial	up to 10	Free, 30 days	Full Starter features
Starter	up to 10	Rp 49k	Rent tracking, WhatsApp reminders, receipts, basic report
Growth	up to 30	Rp 99k	Plus staff accounts, full reports, higher message quota
Pro	up to 100	Rp 199k	Plus multi-property, priority support

- **Annual plan:** pay for 10 months, get 12. Upfront cash helps a solo, self-funded business and lowers churn.
- **Small permanent free tier** (e.g. up to 3-5 rooms) is an option as a growth lever; decide after the pilot.
- **WhatsApp cost:** each plan has a message quota; beyond it, reminders fall back to the share-to-WhatsApp button or are charged.
- **Revenue per owner** = subscription + (tenant fees minus gateway cost). The tenant fee must clear the gateway's current rate card with margin.
- **Later:** add-ons such as digital contracts and utility billing.

Revenue depends on owners paying monthly and tenants choosing online payment, so the product must

clearly save owners time and money.

10. Success Metrics

Metric	Target (pilot)
Activation: owner adds rooms + tenants and sends first invoice within 7 days	≥ 60%
Day-30 retention of pilot owners	≥ 60%
Onboarding time (sign-up to first invoice)	≤ 10 min
On-time payment rate vs owner's previous baseline	+15 pts
Share of invoices paid online (after Phase 2)	≥ 40%
Owner satisfaction (NPS)	≥ 40
Trial-to-paid conversion	≥ 25% (proposed)
Monthly churn of paying owners	≤ 5% (proposed)
MRR	Tracked monthly; target set after pilot pricing
Gross margin per owner (after hosting + WhatsApp cost)	≥ 70% (proposed)
GMV and take rate	Tracked after Phase 2

11. Roadmap

Phase	Duration (est.)	Outcome
0. Validation	2-4 weeks	15-20 owner interviews, manual service pilot with 3-5 owners, pricing signals
1. MVP	10-14 weeks (solo)	Pilot with 10-20 owners, daily use for one month
2. Payments & billing	8-10 weeks (solo)	Online payment, reconciliation, settlement, subscription billing live
3. Growth	Ongoing	Multi-property, staff, complaints, referral program, iOS

12. Risks & Mitigations

Risk	Mitigation
Owners stick to notebooks	Import existing tenants fast; WhatsApp-first; hands-on onboarding in the pilot
Low willingness to pay a subscription	Test prices in the pilot; keep Starter cheap; earn on transactions as a second stream; prove time and money saved
High churn of small owners	Hands-on onboarding, track activation, annual plan discount, show value (reminders sent, money collected)
WhatsApp template rejection or cost	Prepare templates early; fallback share-to-WhatsApp button; budget message costs
Payment/regulatory issues	Use licensed gateways only; legal review before handling settlement
Competition (Mamikos tools, others)	Focus on simplicity and payment collection for small owners, not listings
Data privacy	Minimal ID data, encryption, clear consent
Tenants bypass online payment to avoid the fee	Keep the fee small and the benefit visible (instant receipt, no proof needed); owners can prefer online payment
Solo founder bandwidth	Ship MVP without the payment gateway first; use managed services; defer P1/P2 items
Owners cost money (WhatsApp, hosting) before they pay	Message quota per plan; trial capped at 30 days

13. Decisions & Open Questions

Decisions (v0.3)

#	Topic	Decision
1	Business model	SaaS: owner subscription plus tenant fee on online payments
2	Team	Solo founder; tight MVP scope, managed services
3	Launch market	Jabodetabek (Jakarta, Bogor, Depok, Tangerang, Bekasi); start with one or two clusters
4	Core features	Paid by the owner (subscription), with a free trial
5	Online payment fees	Tenant pays a convenience fee only when paying through the app (pay-if-you-use)
6	Tenants paying owner	No fee

directly

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|---|--------------------|---|
| 7 | Deposit and refund | Owner-selectable policy (see F-26) |
| 8 | Funding | Owner traction first; add funding for infrastructure only if demand is high |
| 9 | Interfaces | Smartphone app for owners, WhatsApp for tenants, internal web admin console |

Still open

1. Plan prices and room tiers; 30-day trial only, or also a small permanent free tier.
2. Exact tenant convenience fee, based on the gateway's current rate card.
3. How owners pay the subscription: payment link each cycle, or recurring auto-debit.
4. First cluster inside Jabodetabek and how to reach the first 20 owners.
5. Legal review of settlement, deposit flows, and UU PDP compliance before Phase 2.
6. Runway: how many months can you work on this before revenue starts?

